

Karl Zhu

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EDUCATION

Massachusetts Institute of Technology

Sep 2023 – May 2028

Candidate for PhD in Operations Research

Cambridge, MA

- GPA: 5.0/5.0.
- Advisor: Prof. Dimitris Bertsimas
- Research areas: adaptive robust optimization for energy systems, reinforcement learning for healthcare.

University of Auckland

Mar 2018 – Nov 2021

Bachelor of Engineering (Honours) in Engineering Science

Auckland, New Zealand

- GPA: 8.7/9.0 (First Class Honours).
- Advisor: Prof. Andy Philpott
- Thesis: Improving multi-stage stochastic programming with distributionally robust optimisation.

WORK EXPERIENCE

Undergraduate R&D Engineer

Nov 2021 – Feb 2022

Beca Engineering

Auckland, New Zealand

- Developed optimization models to improve the energy efficiencies in multiple large cold-storage sites.
- Reduced the 1MW compressors' energy consumption by 5% using machine learning and optimization.

R&D Intern

Nov 2020 – Feb 2021

Fonterra

Auckland, New Zealand

- Stabilized milk allocation decisions by implementing a robust optimization model in Julia.

RESEARCH EXPERIENCE

Research Assistant

Sep 2023 – present

Massachusetts Institute of Technology

Cambridge, MA

- Advisor: Prof. Dimitris Bertsimas
- Optimizing Green Ammonia Investment and Operation. **Methods:** Optimization (adaptive robust, mixed-integer)
- Optimal diagnostic testing for ED differential diagnosis. **Methods:** Reinforcement learning (POMDPs)

Research Assistant

Mar 2022 – Aug 2023

University of Auckland

Auckland, New Zealand

- Supervisors: A/Prof. Andrew Mason, Dr. Anthony Downward
- Data analytics and optimization to find missing rightful shareholders with public data.
- Built an analytics pipeline in Julia that produces a tractable integer program for large-scale inference.

Research Assistant

Nov 2019– May 2020

Photon Factory

Auckland, New Zealand

- Supervisor: Dr. Alex Risos
- Real-time drinking water pathogen detection with ultrafast lasers.
- Developed imaging software in Python, and improved laser beam shaping algorithms with meta-heuristics.

TEACHING EXPERIENCE

Teaching Assistant

Spring – Fall 2025

Massachusetts Institute of Technology

Cambridge, MA

- Courses: *Optimization Methods (15.C57)*, *The Analytics Edge (15.727)*

Teaching Assistant

Mar 2020 – July 2023

University of Auckland

Auckland, New Zealand

- Courses: programming, engineering mechanics, and mathematical modelling.

High School Tutor

Mar 2018 – Oct 2022

MyTuition

Auckland, New Zealand

- For Cambridge International Examinations AS / A Level Math and Physics.

Published / Accepted

1. **Karl K. Zhu**, Andrew J. Mason, Anthony Downward. (2022). Finding the Missing with Integer Programming. *Proc. of the 54th ORSNZ Conference*.
2. Shao, Jing; Wang, Yingming; **Zhu, Karl**; Sun, Zhiyuan; Han, Suli; Dong, Hao; Sun, Shufeng; Song, Ping; Zhang, Ruolan (2024). Interface stress analysis and bonding strengthening exploration of metal layer on the laser-activated copper-clad AlN. *Microelectronics Reliability*, 155.
3. **Karl K. Zhu**, Dimitris Bertsimas. *Overfitting in Adaptive Robust Optimization*. Accepted to the *NeurIPS 2025 Workshop on Machine Learning and Operations Research*. arXiv:2509.16451.

Working Papers

1. **Karl K. Zhu**, Dimitris Bertsimas. Adaptive Robust Optimization for Green Ammonia Capacity Planning and Operations Under Solar and Wind Uncertainty.
2. **Karl K. Zhu**, Yubing Cui, Cindy Wang, Vasiliki Stoumpou, Matthew Peroni, Dimitris Bertsimas. Optimal Diagnostic Test Selection with Finite-Horizon POMDPs.
3. Matthew Peroni, **Karl K. Zhu**, Charles Dai, Joanna Kondylis, Dimitris Bertsimas. (2024). *Brainsage: Learning Meaningful Representations from 3D Brain Scans for Neurological Disease Prediction*.

Conference Abstracts & Presentations

1. Grayson, T., **Zhu, K.**, Amani, R., Aguegaray, C., Broderick, N. G. R., Nieuwoudt, M., & Risos, A. (2020). How much plastic have you drunk today?. *7th Asian Spectroscopy Conference*.
2. **Karl K. Zhu**, Andrew J. Mason, Anthony Downward. (2022). Finding the Missing with Integer Programming. **Best Presentation Award – Young Practitioners’ Prize** (*54th ORSNZ Conference*),
3. Andrew J. Mason, **Karl K. Zhu**, Anthony Downward. (2024). Using integer programming, Markov logic networks and inference to help find missing shareholders in social networks for a New Zealand Māori indigenous incorporation. *EURO 2024*,
4. **Karl Zhu**, Yubing Cui, Cindy Wang, Swathi Rachoor, Nadine Koser, Dimitris Bertsimas. (2024). Automating Care Reviews for Denial Management with Multimodal AI. *INFORMS Annual Meeting*.
5. Matthew Peroni, **Karl Zhu**, Charles Dai, Joanna Kondylis, Dimitris Bertsimas. (2024). Brainsage: Learning Meaningful Representations from 3D Brain Scans for Neurological Disease Prediction. *INFORMS Annual Meeting*.

SKILLS AND ACTIVITIES

Vice President – INFORMS Student Chapter, MIT	2024
Communications Officer – Operations Research Society of New Zealand	2023
Postgraduate Student Representative – Faculty of Engineering, University of Auckland	2022
Treasurer – University of Auckland Brazilian Jiu Jitsu Club	2022
Co-founding President – University of Auckland Chess Association	2019 – 2021
Olympic Trials Candidate – Swimming New Zealand	2016
World Youth Chess Championships – New Zealand team	2008, 2011

CITIZENSHIP

Citizen of New Zealand